

Enterprise Infrastructure Plan

ABC Startup Solutions

Course: ITEP 414 – System Administration and Maintenance
Program: Bachelor of Science in Information Technology
Project: Week 2 Portfolio Project – Enterprise Infrastructure Planning for a Startup Company
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Part 1 — Company Profile

Field	Details
Company Name	ABC Startup Solutions
Nature of Business	Custom software development, web & mobile application development, and IT consulting services for small-to-medium enterprises
Company Vision	To become a leading innovator in software solutions, empowering businesses through reliable, scalable, and secure technology.
Company Mission	To deliver high-quality, secure, and cost-effective software solutions while fostering a culture of continuous learning, collaboration, and technical excellence.
Office Location (fictional)	4th Floor, Innovate Business Center, National Highway, Los Baños, Laguna, Philippines 4030
Total Employees	20

Organizational Structure



Employee Distribution

Department	Employees
Information Technology	5

Human Resources	4
Finance	5
Sales	6
TOTAL	20

Part 2 — Enterprise Hardware Inventory

Asset ID	Hardware	Qty	Department	Purpose
AST-001	Desktop Computer	3	IT	Server administration, helpdesk support, network monitoring
AST-002	Desktop Computer	4	Human Resources	Employee records, payroll processing, recruitment tasks
AST-003	Desktop Computer	5	Finance	Accounting, invoicing, financial reporting
AST-004	Desktop Computer	2	Sales	In-office order processing and CRM data entry
AST-005	Laptop	2	IT	Field troubleshooting, on-site network setup, remote server access
AST-006	Laptop	4	Sales	Client presentations, on-site meetings, remote work
AST-007	Server (Rack-mounted)	2	IT	Primary application/file server + backup/domain controller
AST-008	Router	1	IT (Network Infrastructure)	Routes traffic between the LAN and the ISP connection
AST-009	Managed Switch (24-port)	2	IT (Network Infrastructure)	Connects departmental workstations, servers, and printers
AST-010	Network Printer	2	Shared (HR/Finance/Sales)	Shared document printing and scanning
AST-011	UPS	2	IT (Server Room)	Power backup for the server rack and network equipment
AST-012	Wireless Access Point	2	IT (Network Infrastructure)	Wi-Fi coverage across the single office floor
AST-013	NAS Storage	1	IT	Centralized file storage and automated backup repository
AST-014	External Backup Drive	4	IT/Finance/HR	Offline/offsite backup rotation for critical data
AST-015	Monitor	15	All Departments	Display units for desktop workstations (incl. one dual-monitor setup for IT)

Justification: Device counts mirror the 20-employee headcount (14 desktops + 6 laptops = 20), with laptops reserved for roles requiring mobility (Sales client visits, IT field support). Two servers provide role separation without over-provisioning for a 20-person company. Redundant switches and UPS units protect against single points of failure in the server room.

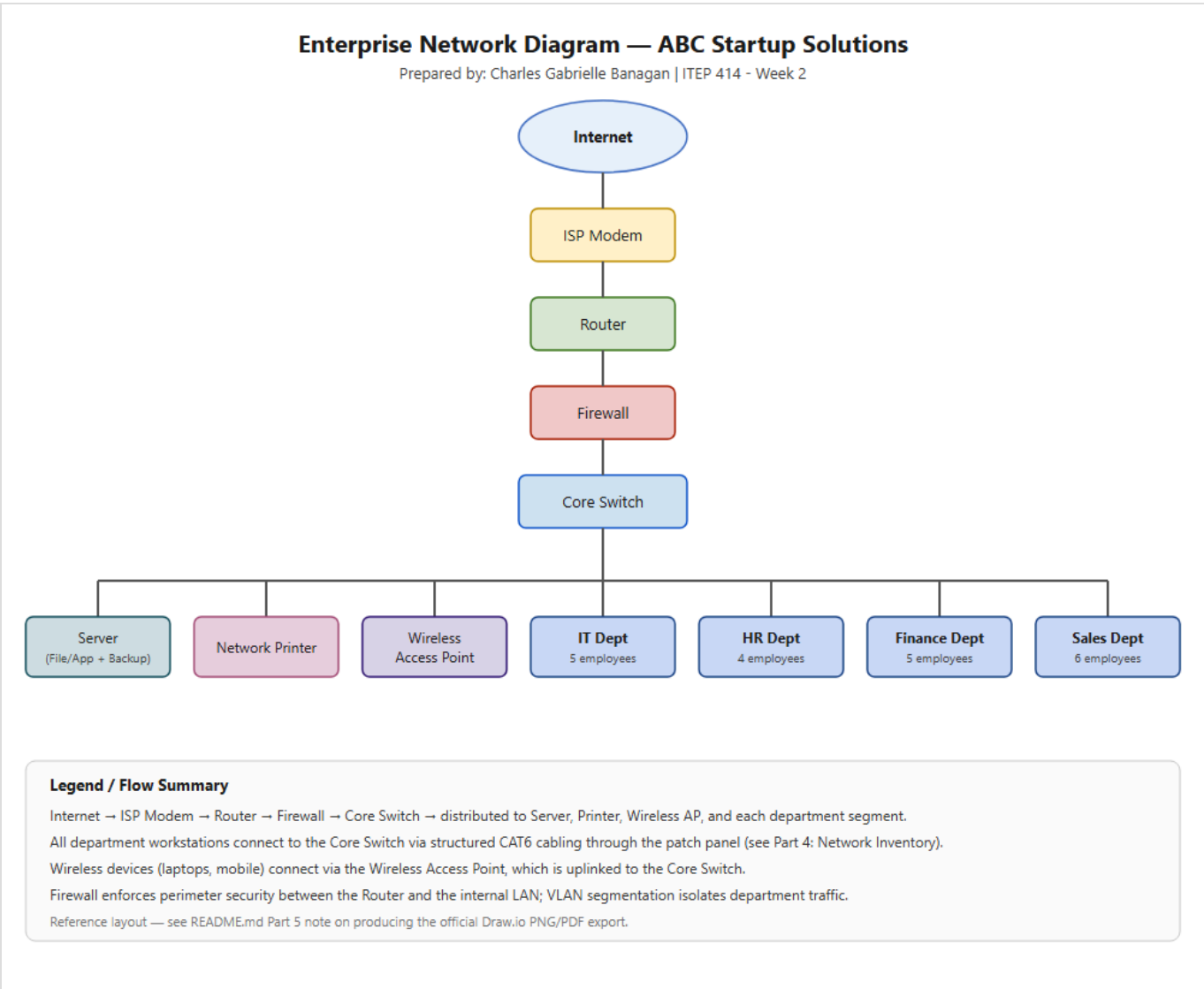
Part 3 — Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume License (per device)	Primary OS for desktops/laptops; supports domain join, BitLocker, Group Policy
Ubuntu Server	22.04 LTS	Open Source (Free)	OS for the primary application/file server — stable, low overhead
Microsoft Office	365 Business Standard	Subscription (per user)	Word processing, spreadsheets, email, collaboration
VS Code	Latest (1.9x)	Free / Open Source	Code editor for IT's scripting and automation
Git	Latest (2.4x)	Free / Open Source	Version control for infrastructure scripts and internal projects
GitHub Desktop	Latest	Free	Simplifies Git repository management
VirtualBox	7.x	Free / Open Source	Virtualization for testing server configurations
Google Chrome	Latest	Free	Primary web browser for daily operations
Microsoft Defender	Built-in (Windows 11)	Included with OS	Endpoint antivirus and malware protection
AnyDesk	Latest	Free / Business License	Remote desktop support tool for the IT helpdesk
7-Zip	23.x	Free / Open Source	File compression and archiving

Part 4 — Enterprise Network Inventory

Asset	Specification	Qty	Purpose
ISP Modem	Fiber modem (up to 500 Mbps)	1	Terminates the ISP fiber line and provides internet connectivity
Router	Enterprise-grade Gigabit router	1	Routes traffic between the internal network and the internet
Firewall	Hardware firewall appliance	1	Filters inbound/outbound traffic, blocks unauthorized access
Switch	24-port managed Gigabit switch	2	Connects wired devices within each department
Access Point	Dual-band wireless AP	2	Wi-Fi coverage for laptops and mobile devices
Patch Panel	24-port CAT6 patch panel	1	Organizes and terminates structured cabling runs
CAT6 Cables	Solid CAT6 UTP cable	300 m	Wired connections between switches, patch panel, workstations
RJ45 Connectors	CAT6-rated RJ45	100 pcs	Terminates CAT6 cable runs into patch panel and wall jacks

Part 5 — Enterprise Network Diagram



Internet → ISP Modem → Router → Firewall → Core Switch → Server / Printer / Wireless AP / IT / HR / Finance / Sales

This diagram was authored as a scalable vector layout and rendered here for the plan document. The companion diagrams/NetworkDiagram.svg, .png, and .pdf files in the repository are the official exports of this topology.

Part 6 — System Administration Roles

Helpdesk Technician

- **Responsibilities:** First-line technical support, hardware/software troubleshooting, ticket management, password/account resets, new-hire device setup.
- **Skills:** Customer service, basic networking, OS troubleshooting (Windows/macOS), clear documentation.
- **Common Tools:** AnyDesk / TeamViewer, Freshservice/Zendesk, Active Directory Users and Computers.
- **Certifications:** CompTIA A+, CompTIA ITF+, Microsoft Certified: Modern Desktop Administrator Associate.

Network Administrator

- **Responsibilities:** Designs and maintains the LAN/WAN, configures routers/switches/firewalls, monitors network performance, manages VPNs and Wi-Fi.
- **Skills:** TCP/IP, VLANs, routing protocols, firewall configuration, network monitoring.
- **Common Tools:** Wireshark, PRTG/SolarWinds, Cisco Packet Tracer, pfSense.
- **Certifications:** CompTIA Network+, Cisco CCNA, CompTIA Security+.

Linux System Administrator

- **Responsibilities:** Manages Linux servers, user/permission management, package and service management, automates tasks with shell scripts, applies security patches.
- **Skills:** Bash scripting, systemd, file permissions, package managers (apt/yum), cron scheduling.
- **Common Tools:** SSH, Bash, Ansible, Nagios/Zabbix.
- **Certifications:** LFCS, RHCSA, CompTIA Linux+.

Cloud Administrator

- **Responsibilities:** Manages cloud infrastructure (IaaS/PaaS), provisions VMs and storage, manages cloud backups, optimizes cost, administers cloud IAM.
- **Skills:** Cloud networking, identity and access management, infrastructure-as-code (Terraform), cost monitoring.
- **Common Tools:** AWS Console / Azure Portal / GCP Console, Terraform, CloudWatch / Azure Monitor.
- **Certifications:** AWS Certified Solutions Architect – Associate, Microsoft Certified: Azure Administrator Associate, Google Associate Cloud Engineer.

How They Work Together

At ABC Startup Solutions, these four roles form a layered support system rather than working in isolation. The **Helpdesk Technician** is the first point of contact for day-to-day employee issues and escalates anything beyond basic troubleshooting. The **Network Administrator** ensures the underlying connectivity — routers, switches, firewall, and Wi-Fi — stays available and secure, which every other role depends on. The **Linux System Administrator** keeps the on-premises server (file storage, backups, internal services) patched, running, and properly permissioned. As the company grows and adopts cloud services for scalability, the **Cloud Administrator** extends that same reliability and security discipline to hosted infrastructure, working closely with the Network Administrator on VPN/hybrid connectivity and with the Linux Administrator on server migration and backup strategy. Together, they cover the full stack from end-user support to network, server, and cloud — with clear escalation paths preventing any single point of failure in the company's IT support model.

Part 7 — Infrastructure Recommendations

- **Internet Provider:** A business-grade fiber ISP (e.g., PLDT Enterprise or Converge ICT Business) with an SLA-backed connection of at least 300–500 Mbps symmetrical, plus a secondary LTE/backup line for failover during outages.
- **Server Specifications:** A rack-mounted server in the Dell PowerEdge T340/R340 class — Intel Xeon E-2300 series CPU, 32–64 GB ECC RAM, RAID 1/5 across 2–4 SSDs, redundant power supply — running Ubuntu Server 22.04 LTS, virtualized to host file server, domain services, and backup roles.
- **Backup Strategy:** Follow the 3-2-1 rule — 3 copies of data, on 2 different media types (NAS + external drive), with 1 copy offsite/cloud. Automated nightly incremental backups plus a weekly full backup.
- **Security Recommendations:** Hardware firewall with IDS/IPS, VLAN segmentation per department, VPN for remote access, scheduled patch management, periodic employee security-awareness training.
- **Antivirus:** Microsoft Defender for Business, centrally managed, supplemented by scheduled full-system scans across all endpoints.
- **Password Policy:** Minimum 12 characters combining uppercase, lowercase, numbers, and symbols; mandatory rotation every 90 days; MFA required for email and admin accounts; account lockout after 5 failed attempts; no reuse of the last 5 passwords.
- **Expansion Plan:** Provision the network with a /24 subnet (254 usable addresses) and ~20% spare switch ports; leave modular rack space in the server room; size Wi-Fi AP coverage for future desks; design with a hybrid-cloud path in mind for growth past ~50 employees.

Part 8 — Personal Reflection

Working through this Enterprise Infrastructure Plan gave me a much more concrete sense of what "planning" actually means in system administration. Going in, I assumed the hard part of setting up a company's IT would be the technical configuration — installing servers, wiring switches, setting up firewalls. What this project showed me instead is that the hardest and most important work happens *before* any of that: understanding how many people are in each department, what they actually do all day, and translating that into a realistic list of hardware, software, and network equipment that the business can justify buying.

The most challenging part for me was the hardware and network inventory. It's easy to just list "20 desktops, 1 server, 1 router" and call it done, but the assignment pushed me to actually reason about *why* each department needed what it needed — why Sales gets laptops instead of desktops, why the server room needs redundant UPS units, why a 20-person company still benefits from VLAN segmentation. Justifying those choices forced me to think like the Junior System Administrator described in the scenario, not just like a student filling out a table.

Designing the network diagram was the second major challenge. Mapping the logical flow from the internet down through the modem, router, firewall, and switch, and then out to each department, made me appreciate how a badly planned network topology can create bottlenecks or security gaps that are invisible until something breaks. It also reinforced why documentation matters — a diagram that only makes sense to the person who drew it isn't useful to a team.

This project drove home why planning has to come before deployment. If ABC Startup Solutions bought hardware first and figured out the network second, they'd likely end up with mismatched equipment, insufficient cabling, or a network that can't scale past its first year. Planning lets a System Administrator catch those problems on paper, where they're cheap to fix, instead of after the equipment is already installed.

Overall, this exercise pushed me to think beyond individual tools and toward infrastructure as a system — hardware, software, network, security policy, and people all depending on each other. That systems-level thinking, plus the practice of documenting technical decisions clearly enough for a non-technical manager to approve them, is exactly the skill set I'll need whether I end up as a Network Administrator, a Linux Administrator, or a Cloud Administrator down the line.